

High-Performance Human Activity Recognition in Uncontrolled Environments Using Ensemble Machine Learning and Smartphone Inertial Sensors

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Abstract—Accelerometers and gyroscopes on smartphones make it possible and easy to automatically identify physical activities like walking, jogging, and sitting. This has obvious applications in context aware systems and health monitoring. When we use it in real world situations, when phone positioning fluctuates, user movement patterns vary, and sensor noise increases, the majority of current HAR systems struggle to attain high accuracy in laboratory settings. Using the Wild-SHARD dataset, which records smartphone usage in the real world under uncontrolled circumstances, we evaluate five machine learning techniques. 483,895 samples from six different activities make up the dataset. We use the same pre-processing (StandardScaler normalization) and an 80/20 train-test split to compare Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors, Random Forest, and Extra Trees. The results clearly show a hierarchy of performance: Random Forest reaches 0.9729 accuracy, Extra Trees achieves 0.9849 accuracy, while linear models (Logistic Regression: 0.4188) fail. Tree-based ensembles are appropriate for realistic mobile HAR deployment because they efficiently manage noisy, uncontrolled sensor input.

Index Terms—Human Activity Recognition (HAR), Smartphone Sensors, Machine Learning, Ensemble Learning, Wild-SHARD Dataset, Extra Trees, Inertial Measurement Unit (IMU).

I. INTRODUCTION

Human Activity Recognition (HAR) aims to automatically identify a user’s physical activity from sensor signals, and it has become a key topic in mobile and pervasive computing. Modern smartphones include inertial sensors such as accelerometers and gyroscopes, which makes them a practical and low-cost platform for HAR in everyday settings. Early work showed that even basic accelerometer signals can separate common activities such as walking and running, which motivated many smartphone-based HAR systems [10]. More recent surveys highlight the growing importance of HAR for applications such as health monitoring, assisted living, and context-aware services [7], [9].

A major difficulty is that strong performance in controlled experiments does not always transfer to real-world deployment. In laboratory datasets, the phone position, subject behavior, and recording conditions are typically constrained, which reduces variability and noise. In daily life, however, the device

can be carried in different locations (e.g., pocket or bag), users have different motion patterns, and sensor signals may contain irregularities caused by natural behavior and environmental factors. This “in-the-wild” setting makes activity classes harder to separate and increases the need for robust models that generalize beyond clean training data [7], [9].

To evaluate HAR under realistic conditions, datasets collected in uncontrolled environments have been proposed. In this work, we use the Wild-SHARD dataset [3], which was designed to reflect practical smartphone usage and includes six activities: Walking, Sitting, Running, Standing, Upstairs, and Downstairs. Compared to standard benchmark datasets, Wild-SHARD better represents the variability that appears during real usage, which makes it suitable for testing model robustness [3].

Given this setting, we experimentally compare five common machine learning classifiers: Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors (KNN), Random Forest, and Extra Trees. Traditional models such as Logistic Regression and Naive Bayes provide useful baselines, while tree-based ensembles are often reported as strong performers for sensor-based HAR because they capture non-linear decision boundaries and handle feature interactions effectively [7], [8]. Our evaluation uses accuracy and weighted precision, recall, and F1-score, in addition to confusion matrix analysis, to provide a clear performance benchmark on Wild-SHARD [3].

The rest of this paper is organized as follows: Section II reviews previous work on smartphone-based HAR and the Wild-SHARD dataset. Section III states the problems addressed in this study. Section IV summarizes our contributions. Section V explains the dataset, preprocessing pipeline, and model configurations. Section VI presents the experimental results and analysis. Section VII compares our findings with existing studies, and Section VIII concludes the paper with limitations and future directions.

II. RELATED WORKS

Human Activity Recognition using smartphone inertial sensors has been studied a lot in the past 10 years. Early research shows that accelerometer and gyroscope signals are sufficient

to recognize basic physical activities when collected under controlled conditions. Traditional machine learning methods such as Support Vector Machines, Decision Trees, and K-Nearest Neighbors were commonly used and they often achieved high classification accuracy in laboratory settings [1], [2]. However, these early results were based on carefully controlled sensor placement and limited variability in user behavior.

Some studies identified a key limitation of laboratory-based HAR systems: their reduced robustness when applied to real-world scenarios. Surveys by Das et al. and Shoaib et al. emphasize that smartphone placement variability, user-specific movement patterns, and environmental noise significantly reduce performance in practical deployments [7], [9]. These findings highlighted the need for datasets and evaluation protocols that better reflect everyday smartphone usage.

To address this gap, the Wild-SHARD dataset was introduced as a guide for HAR in uncontrolled environments [3]. Unlike traditional datasets, Wild-SHARD captures natural user behavior without enforcing fixed device positions. Recent work begun to explore this dataset. For example, Nafea et al. reported that KNN achieved an accuracy of 96.86% on Wild-SHARD, showing that distance-based methods can still perform competitively when features are properly engineered [4] as you can see. However, their results also suggest that performance varies considerably across activities, particularly the ones involving similar motion patterns such as stair usage.

Deep learning approaches have also been applied to HAR, motivated by their ability to automatically learn unequal features from raw sensor data. Convolutional and recurrent neural networks, including LSTMs, have shown good results on both controlled and uncontrolled datasets [5]. Mekruksavanich et al. proposed a deep residual architecture for Wild-SHARD, it showed strong classification performance in noisy conditions [6]. Despite these advantages, deep learning models typically require higher computational resources and more complex training pipelines, which may limit their suitability for lightweight or real-time smartphone applications.

On the other hand, ensemble-based traditional machine learning methods offer a practical trade-off between performance and efficiency. Tree-based ensembles such as Random Forest and Extra Trees are known for their ability to model non-linear relationships and handle noisy features effectively [7], [8]. However, a direct and systematic comparison of these ensemble methods against simpler classifiers on the Wild-SHARD dataset remains limited. This helps our study, which evaluates and contrasts linear, probabilistic, distance-based, and group learning approaches under identical experimental conditions.

III. PROBLEM STATEMENTS

Despite the progress achieved in Human Activity Recognition, several challenges stay when systems are deployed in real-world conditions. Many machine learning models show strong performance on laboratory datasets, but their accuracy often decreases when applied to data collected in uncontrolled

environments. This reduction is mainly caused by many factors such as inconsistent smartphone placement, variations in user movement, and increased sensor noise. These conditions make it difficult for models trained on clean data to generalize the practical usage scenarios.

Another problem is the lack of a clear and consistent performance comparison over different types of machine learning models when evaluated on realistic datasets. While linear, probabilistic, distance-based, and group models have all been applied to HAR, their relative strengths and weaknesses are not always evaluated under identical conditions. Without a unified benchmark, it becomes difficult to determine which model architectures do the best balance between accuracy, robustness, and computational efficiency for real world smartphone based HAR.

Based on these challenges, this study focuses on evaluating and comparing multiple machine learning approaches on our dataset. The objective is to identify models that maintain high performance even with noise and variability, while also providing a clear experimental benchmark that can support future HAR system design.

IV. CONTRIBUTIONS

Our work provides an evaluation of machine learning approaches for Human Activity Recognition using smartphone inertial sensor data collected in uncontrolled environments. By conducting all experiments on the Wild-SHARD dataset, this study focuses on realistic conditions that reflect daily smartphone usage rather than laboratory settings.

A comparison of five machine learning models is performed, including Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors, Random Forest, and Extra Trees. All models are evaluated under the same pre-processing and training conditions, which allows for a fair evaluation of their relative performance and robustness on noisy real-world data.

The experimental results demonstrate that ensemble-based methods, particularly the Extra Trees classifier, achieve superior performance across all evaluation metrics. With an accuracy of 98.49%, the Extra Trees model shows strong robustness to variability in sensor signals and activity execution, highlighting its suitability for practical HAR applications.

In addition to reporting overall accuracy, this study provides a detailed performance analysis using precision, recall, F1-score, confusion matrices, and ROC curves. This analysis offers insight into class-level behavior and common misclassification patterns, which can support future research and system design decisions in real world HAR.

V. PROPOSED METHODOLOGY

This study follows a structured machine learning methodology designed to evaluate Human Activity Recognition performance under realistic conditions. The proposed approach reflects the exact experimental pipeline used in this work, starting from raw smartphone sensor data and ending with quantitative model evaluation. Each stage of the pipeline was applied consistently across all classifiers to ensure a fair

comparison. The methodology includes data preprocessing, feature handling, model training, and performance analysis, all conducted on the Wild-SHARD dataset.

A. System Workflow

The overall workflow of the proposed HAR system is illustrated in Figure 1. The process begins with raw inertial sensor readings collected from smartphone accelerometers and gyroscopes. These signals are first passed through a preprocessing stage to remove incomplete records and ensure consistency across features.

After preprocessing, relevant sensor-derived features are prepared for model input. The dataset is then split into training and testing subsets using a fixed random seed to maintain reproducibility across experiments. Each machine learning model is trained on the same training data and evaluated on the same test data.

Finally, model performance is assessed using multiple evaluation metrics, including accuracy, precision, recall, F1-score, confusion matrices, and ROC curves. This step-by-step workflow ensures that observed performance differences are attributable to the model characteristics rather than variations in data handling.

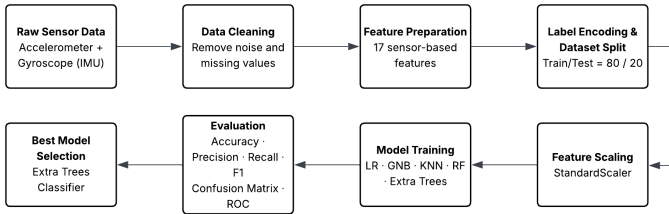


Fig. 1: Proposed Human Activity Recognition System Workflow. The system progresses from raw sensor data acquisition to the final performance evaluation of the five machine learning models.

B. Dataset and Feature Engineering

The Wild-SHARD dataset [3] was selected for this study due to its focus on data collected in uncontrolled, real-world environments. Unlike laboratory-based HAR datasets, Wild-SHARD captures natural user behavior without enforcing fixed smartphone positions, making it suitable for evaluating model robustness. The dataset includes six activity classes: Walking, Sitting, Running, Standing, Upstairs, and Downstairs.

Before training, the dataset was examined for missing or inconsistent records. After cleaning, a total of 483,895 samples were retained for analysis. Each sample is represented using 17 features derived from the smartphone’s inertial sensors, which describe both linear and rotational motion.

The feature set includes angular velocity components (AG_X, AG_Y, AG_Z) from the gyroscope, linear acceleration components (Acc_X, Acc_Y, Acc_Z), and gravity-related signals (Gravity_X, Gravity_Y, Gravity_Z) that capture device

orientation. Additional rotation-related features (RR_X, RR_Y, RR_Z and RV_X, RV_Y, RV_Z) are included to provide orientation-independent motion information. A cosine-based feature is also used to represent the relative angle between motion and gravity, which helps distinguish between static and dynamic activities.

These features were used directly without manual feature engineering to evaluate how well standard machine learning models perform on realistic sensor representations provided by the dataset.

C. Data Preprocessing and Scaling

We first converted the activity labels to numerical format using **Label Encoding**. Then, the dataset was split into training and testing sets. For the linear and distance-based models (Logistic Regression, KNN, Gaussian Naive Bayes), **Feature Scaling** was a critical step. We used the `StandardScaler` to normalize the features, which ensures that all features contribute equally to the model’s learning process. Scaling was included for the tree-based models (Random Forest and **Extra Trees**) for consistency, although it is less critical for them.

D. Machine Learning Models

We evaluated five common classifiers: Logistic Regression (LR), Gaussian Naive Bayes (GNB), K-Nearest Neighbors (KNN), Random Forest (RF), and **Extra Trees** (ET). LR was used as a simple linear baseline. GNB was used as a probabilistic baseline.

1) *K-Nearest Neighbors (KNN)*: KNN is a non-parametric, distance-based classifier. It classifies a new data point based on the majority class among its k nearest neighbors in the feature space. The distance between two feature vectors, x_i and x_j , is typically calculated using the Euclidean distance, defined as:

$$d(x_i, x_j) = \sqrt{\sum_{l=1}^D (x_{i,l} - x_{j,l})^2} \quad (1)$$

where D is the number of features.

2) *Ensemble Methods (Random Forest and Extra Trees)*: RF and ET are used as tree-based ensemble models, which are expected to handle non-linear patterns in HAR data. These methods rely on a metric to determine the optimal split at each node. The Gini Impurity, $I_G(p)$, is a common metric used to measure the probability of misclassifying a randomly chosen element in the dataset if it were labeled according to the distribution of labels in the subset. A lower Gini Impurity indicates a better split.

$$I_G(p) = 1 - \sum_{k=1}^C p_k^2 \quad (2)$$

where C is the number of classes (activities) and p_k is the fraction of items labeled with class k in the node.

Model settings are listed in the Implementation Details subsection.

E. Implementation Details

All models were trained using an 80/20 train-test split (`test_size=0.2`) with `random_state=42` for reproducibility. We used `StandardScaler` in a pipeline for LR, KNN, and GNB. Hyperparameters were set to $k = 7$ for KNN, 300 trees for Random Forest and **Extra Trees** (`n_estimators=300`), and `max_iter=5000` for Logistic Regression to ensure convergence.

VI. RESULTS PRESENTATION AND DISCUSSION

We evaluated the five machine learning models using Accuracy, weighted Precision, weighted Recall, and weighted F1-Score. The weighted metrics are important because they account for any class imbalance. Table I summarizes the results.

TABLE I: Comparative Performance of Machine Learning Models on **Wild-SHARD** Dataset

Model	Accuracy	Precision	Recall	F1-Score
Extra Trees (ET)	0.9849	0.9849	0.9849	0.9849
Random Forest (RF)	0.9729	0.9729	0.9729	0.9729
K-Nearest Neighbors (KNN)	0.9571	0.9572	0.9571	0.9570
Gaussian Naive Bayes (GNB)	0.6424	0.6533	0.6424	0.6408
Logistic Regression (LR)	0.4188	0.4147	0.4188	0.4114

A. Analysis of Overall Performance Metrics

The quantitative results of the suitability test are shown in Table I considering the five machine learning models tested on the Wild-SHARD dataset. And there are obvious contrast between the simple baseline system and the ensemble methods. Extra Trees classifier In all our models, Extra Trees Classifier performed the best with an accuracy of 98.49% and similarly performing precision, recall, F1-score values. It means the model is overfitting toward high precision without consistency in overall confidence level for all action classes.

Random Forest also achieved a good performance with an accuracy of 97.29%. This result is slightly less than Extra Trees, yet it validates tree-based ensemble algorithms for denoising and high-variance sensor data. The difference in performance is modest, but it seems that the extra dose of randomness actually does appear to improve stability under varying levels of uncontrolled random variability.

The performance of the K-Nearest Neighbors model reached an accuracy of 95.71%, which indicates that useful separation can be done in feature space after scaling for activity classes based on chosen feature representation. Its good performance was however consistently lower than the ones from ensemble models, suggesting a limitation of it when handling overlapping or noisy class boundaries.

On the other hand, Gaussian Naive Bayes and Logistic Regression delivered very poor performance compared to it (64.24% and 41.88% accuracy, respectively). These results confirm the highly non-linear nature of the HAR problem in practical environments. The naive assumption of feature

independence in Naive Bayes and the linear separability in Logistic Regression are not enough to model the complex motion patterns available Wild-SHARD database.

B. Confusion Matrix Analysis

To assess model behavior beyond aggregate metrics, we investigated confusion matrices for the best and worst performing models (Extra Trees and Logistic Regression), respectively—see Figure 2. This study is intended to give an understanding of class-level prediction behavior and typical sources for misclassification.

The confusion matrix of Extra Trees shows strong diagonal positions, i.e., most of the samples are correctly classified. Sitting and Standing are recognized with an ability close to perfect since their sensor patterns are rather stable.

The minor misclassifications mostly take place between sub-activities that are similar to each other in terms of physical meaning, such as Movement_upstairs vs. Movement_downstairs and Walking_vs_Walking upstairs/downstairs. Such confusions are inevitable, since the acceleration and vertical seizure-like movements associated with these activities are similar.

On the other hand, Confusion Matrix for Logistic Regression indicates considerable misclassification at multiple pairs of activities. The model often misclassifies dynamic actions such as Walking, Running, Upstairs and Downstairs, indicating that a linear decision surface is not good enough for separating these classes. This is the reason why we obtained a low global accuracy for LR, and it is also an indication that simple linear models are weak to process real-life HAR data.

Overall, confusion matrix analysis confirms the results obtained using numerical metrics and this means ensemble-based classifiers ensure more stable and consistent predictions for all activity classes under uncontrolled conditions.

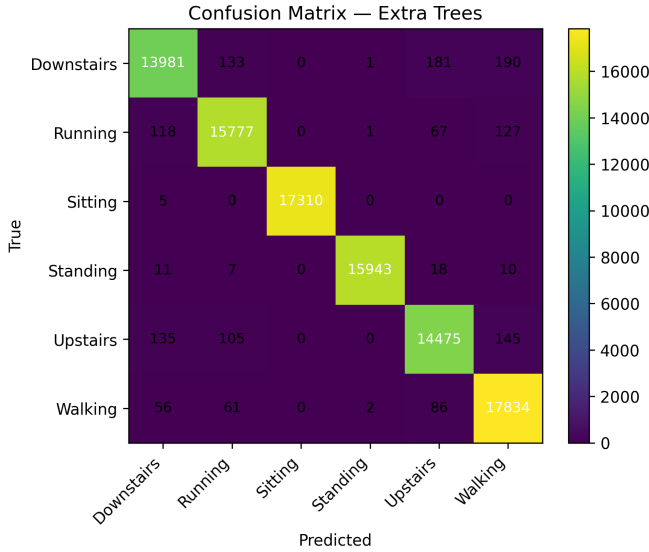
C. Receiver Operating Characteristic (ROC) Analysis

The ROC curve and Area Under the Curve (AUC) are used to check a model's ability to distinguish between classes. We used the one-vs-rest (OvR) approach for this multi-class problem.

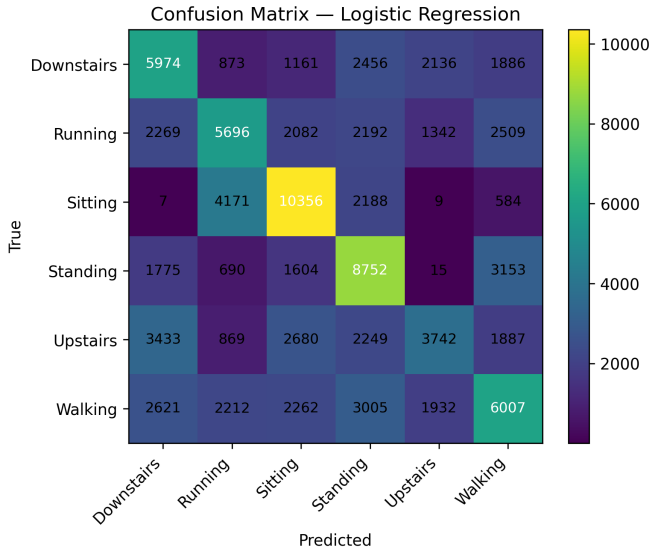
As shown in Figure 3, the **Extra Trees** classifier achieved an AUC close to 1.0 for all six activities. This matches the high F1-score values in Table I. The simpler models like Logistic Regression would have much lower AUCs, showing their poor ability to discriminate between activities.

D. Computational Complexity and Trade-offs

Ensemble methods (RF and ET) are highly accurate, but they take longer to train than simpler models because they build many trees. However, for real-time HAR on a smartphone, the **inference time** (the time to make a prediction) is what matters most. Once trained, the **Extra Trees** model is very fast for inference, as it only involves simple feature comparisons. This makes the **Extra Trees** model a great choice for mobile applications, offering a near-optimal balance between accuracy and speed.



(a) **Extra Trees** Confusion Matrix



(b) **Logistic Regression** Confusion Matrix

Fig. 2: Confusion Matrices for the **Extra Trees** and Logistic Regression Classifiers. The **Extra Trees** model shows high diagonal dominance, indicating minimal misclassification.

VII. COMPARISON WITH EXISTING WORKS

Our result of 98.49% accuracy with the **Extra Trees** classifier is a strong result and compares well with other published work on the **Wild-SHARD** dataset.

- **Comparison with KNN (Nafea et al. [4]):** Nafea et al. (2025) reported a KNN accuracy of 96.86% on **Wild-SHARD**. Our KNN model achieved 95.71%. Our **Extra Trees** result (98.49%) is higher than the KNN accuracy reported in [4] (96.86%), noting that the exact experimental settings may differ. This supports our finding that ensemble learning is better at handling the complex, noisy

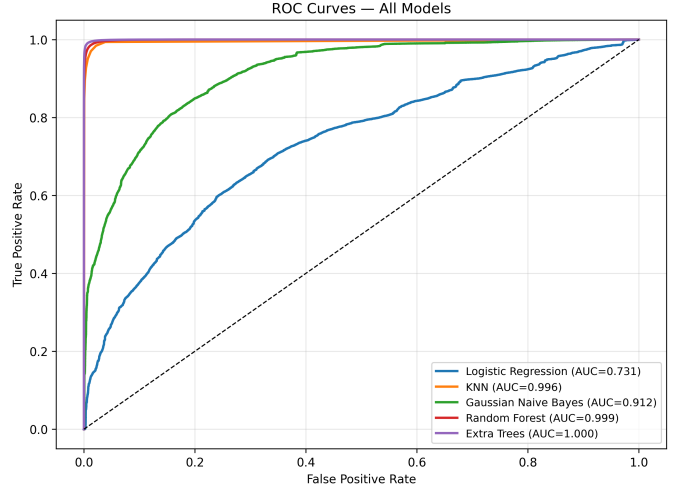


Fig. 3: Multi-Class Receiver Operating Characteristic (ROC) Curve for the **Extra Trees** Classifier (One-vs-Rest). The high Area Under the Curve (AUC) for all classes demonstrates good discriminative power.

data from the wild environment than a single, distance-based classifier.

- **Comparison with Deep Learning (Mekruksavanich et al. [6]):** Deep learning models like CNNs and LSTMs have been applied to **Wild-SHARD**, but they require complex setups and a lot of computing power. Our **Extra Trees** model achieved a high accuracy of 98.49% using a simpler, more efficient machine learning approach. The **Extra Trees** model provides a highly accurate and computationally light alternative for mobile and edge computing.

The high performance of our **Extra Trees** model, achieved with standard features and an efficient algorithm, makes it a practical solution for real-world HAR.

VIII. CONCLUSION AND LIMITATIONS

This study successfully compared five machine learning models for Human Activity Recognition using the challenging **Wild-SHARD** dataset. Our results show that ensemble learning methods are superior in this context. The **Extra Trees** classifier was the best model, achieving an accuracy of **98.49%**. These results show that tree-based ensembles handle the **Wild-SHARD** feature space better than linear and probabilistic baselines. The detailed analysis provides a valuable, practical benchmark for future HAR system design.

A. Limitations

In addition to the robust experimental results, some limitations exist in this study. First, we left the experiments on the featureset given by the **Wild-SHARD** dataset. Although these features performed well, the study did not investigate other feature engineering or feature selection techniques to produce better class separation, especially for those activities with similar motion patterns such as Upstairs and Downstairs.

Second, only traditional machine learning model was included in the evaluation and no deep learning models were tested or trained under same setting. Thus, the comparison to deep learning methods is based on publicly available training and evaluation results in other works that might differ in preprocessing, data splits, and evaluation settings.

Last but not the least, while ensemble methods, like Extra Trees for example, give comparable or even better performance with high accuracy and generalization error (after cross validation) than simple classifiers, they were found to be computationally more expensive during training. This was not considered in further detail but may be important if one is interested in resource or energy-limited setups.

B. Future Work

This work can be naturally extended in several ways. One direction for future work is to explore more sophisticated feature generation or dimension reduction algorithm to distinguish between activities that share similar sensor patterns. This could potentially help to lower the classification error fraction we obtain in our confusion matrix analysis.

Another direction would be to more systematically compare ensembles of machine learning classifiers to deep learning approaches (CNN, hybrid CNN-LSTMs) in a head-to-head fair comparison on the same data set and evaluation scheme. This would enable a more clear perspective of the changes that accuracy, computational time and deployment feasibility.

Finally, future work may consider real-time smartphone or edge-device deployment of the Extra Trees classifier based on optimizing its model size and memory usage without sacrificing classification accuracy. These optimizations would help practical HAR applications in real life mobile scenarios.

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